

Signals & System

Lecture 18

Discrete-Time Transfer Functions

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Discrete-Time Transfer Functions

Transfer Function of a Discrete-Time Systems (2 lectures): Impulse sampler, **Laplace transform of impulse sequence, z transform.** Properties of the z transform. Examples. Difference equations and differential equations. Digital filters.

Specific objectives for today:

- z-transform of an impulse response
- z-transform of a signal
- Examples of the z-transform

Lecture 18: Resources

Core material

SaS, O&W, C10

Related Material

MIT lecture 22 & 23

The z-transform of a discrete time signal closely mirrors the Laplace transform of a continuous time signal.

Reminder: Laplace Transform

The continuous time Laplace transform is important for two reasons:

- It can be considered as a Fourier transform when the signals had infinite energy
- It decomposes a signal $x(t)$ in terms of its **basis functions** e^{st} , which are only altered by magnitude/phase when passed through a LTI system.

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st} dt$$

Points to note:

- There is an associated **Region of Convergence**
- Very useful due to definition of **system transfer function** $H(s)$ and performing convolution via multiplication
 $Y(s)=H(s)X(s)$

Discrete Time EigenFunctions

Consider a discrete-time input sequence (z is a complex number):

$$x[n] = z^n$$

Then using discrete-time convolution for an LTI system:

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$$

$$= \sum_{k=-\infty}^{\infty} h[k]z^{n-k}$$

$$= z^n \sum_{k=-\infty}^{\infty} h[k]z^{-k}$$

$$= H(z)z^n = H(z)x[n]$$

Z-transform of the impulse response

$$H(z) = \sum_{k=-\infty}^{\infty} h[k]z^{-k}$$

But this is just the input signal multiplied by $H(z)$, the **z-transform of the impulse response**, which is a complex function of z .

z^n is an **eigenfunction of a DT LTI system**

z-Transform of a Discrete-Time Signal

The **z-transform** of a discrete time signal is defined as:

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

This is analogous to the CT Laplace Transform, and is denoted:

$$x[n] \xleftrightarrow{Z} X(z)$$

To understand this relationship, put z in **polar coords**, i.e. $z=re^{j\omega}$

$$\begin{aligned} X(re^{j\omega}) &= \sum_{n=-\infty}^{\infty} x[n](re^{j\omega})^{-n} \\ &= \sum_{n=-\infty}^{\infty} (x[n]r^{-n})e^{-j\omega n} \end{aligned}$$

Therefore, this is just equivalent to the scaled **DT Fourier Series**:

$$X(re^{j\omega}) = F\{x[n]r^{-n}\}$$

Geometric Interpretation & Convergence

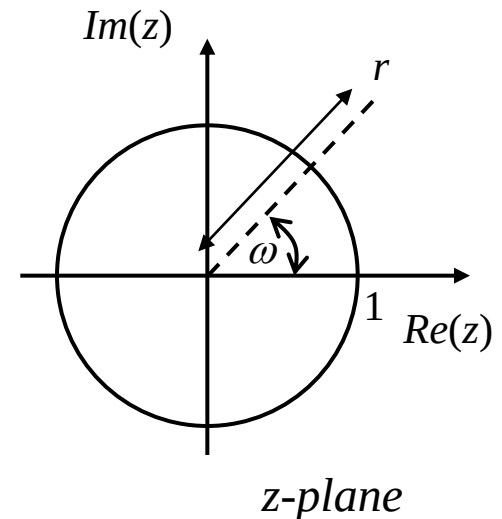
The relationship between the z-transform and Fourier transform for DT signals, closely parallels the discussion for CT signals

The z-transform reduces to the DT Fourier transform when the magnitude is unity $r=1$ (rather than $\text{Re}\{s\}=0$ or purely imaginary for the CT Fourier transform)

For the z-transform convergence, we require that the Fourier transform of $x[n]r^n$ converges. This will generally converge for some values of r and not for others.

In general, the z-transform of a sequence has an associated range of values of z for which $X(z)$ converges.

This is referred to as the Region of Convergence (ROC). If it includes the unit circle, the DT Fourier transform also converges.



Example 1: z-Transform of Power Signal

Consider the signal $x[n] = a^n u[n]$

Then the z-transform is:

$$X(z) = \sum_{n=-\infty}^{\infty} a^n u[n] z^{-n} = \sum_{n=0}^{\infty} (az^{-1})^n$$

For convergence of $X(z)$, we require

$$\sum_{n=0}^{\infty} (az^{-1})^n < \infty$$

The region of convergence (ROC) is

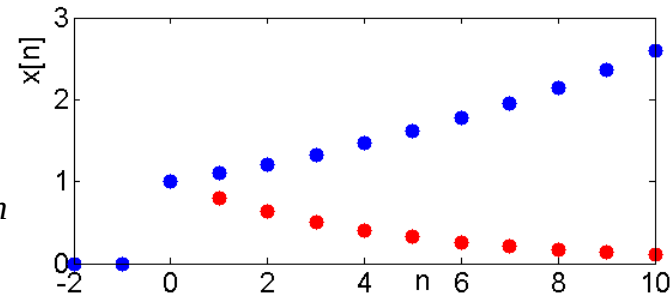
$$|az^{-1}| < 1 \quad \text{or} \quad |z| > |a|$$

and the Laplace transform is:

$$X(z) = \sum_{n=0}^{\infty} (az^{-1})^n = \frac{1}{1 - az^{-1}} = \frac{z}{z - a}, \quad |z| > |a|$$

When $x[n]$ is the unit step sequence $a=1$

$$X(z) = \frac{1}{1 - z^{-1}}, \quad |z| > 1$$



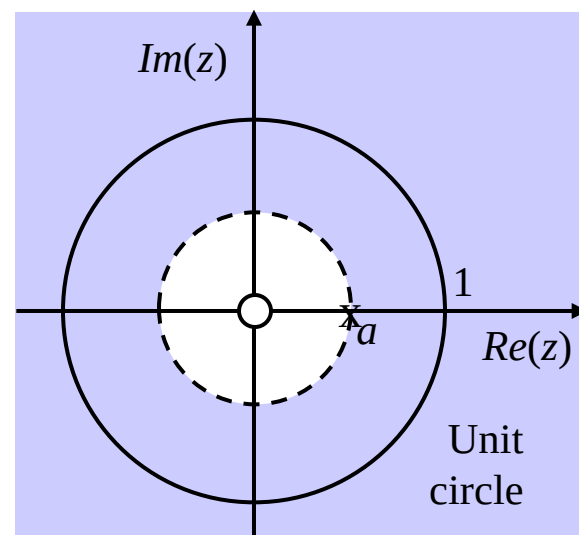
Example 1: Region of Convergence

The z-transform $X(z) = z/(z-a)$ is a rational function so it can be characterized by its **zeros** (numerator polynomial roots) and its **poles** (denominator polynomial roots)

For this example there is one zero at $z=0$, and one pole at $z=a$.

The pole-zero and ROC plot is shown here

For $|a|>1$, the ROC does not include the unit circle, for those values of a , the discrete time Fourier transform of $a^n u[n]$ does not converge.



Example 2: z-Transform of Power Signal

Now consider the signal $x[n] = -a^n u[-n-1]$

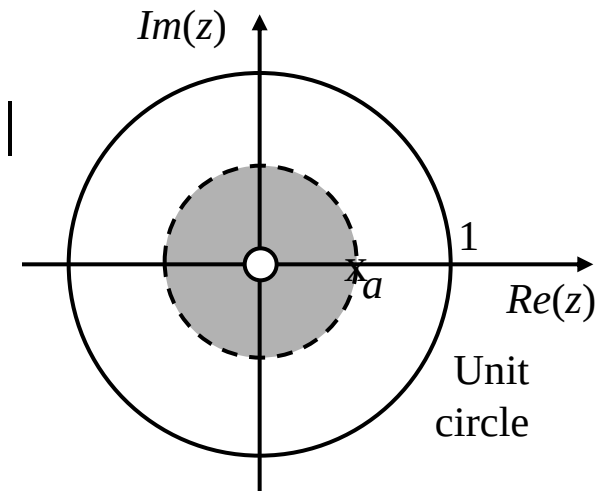
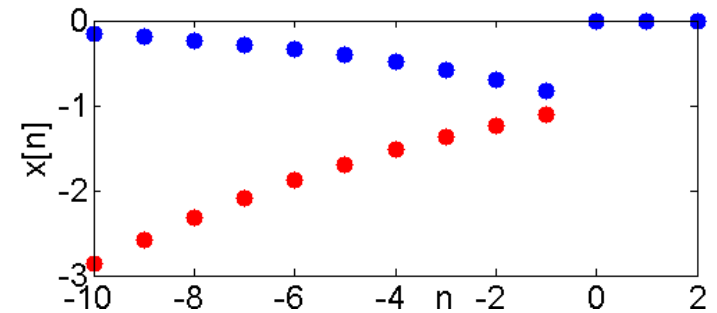
Then the Laplace transform is:

$$\begin{aligned} X(z) &= -\sum_{n=-\infty}^{\infty} a^n u[-n-1] z^{-n} = -\sum_{n=-\infty}^{-1} a^n z^{-n} \\ &= -\sum_{n=1}^{\infty} a^{-n} z^n = 1 - \sum_{n=0}^{\infty} (a^{-1} z)^n \end{aligned}$$

If $|a^{-1}z| < 1$, or equivalently, $|z| < |a|$, this sum converges to:

$$X(z) = 1 - \frac{1}{1 - a^{-1}z} = \frac{1}{1 - az^{-1}} = \frac{z}{z - a}, \quad |z| < |a|$$

The pole-zero plot and ROC is shown right for $0 < a < 1$



Example 3: Sum of Two Exponentials

Consider the input signal

$$x[n] = 7(1/3)^n u[n] - 6(1/2)^n u[n]$$

The z-transform is then:

$$X(z) = \sum_{n=-\infty}^{\infty} \{7(1/3)^n u[n] - 6(1/2)^n u[n]\} z^{-n}$$

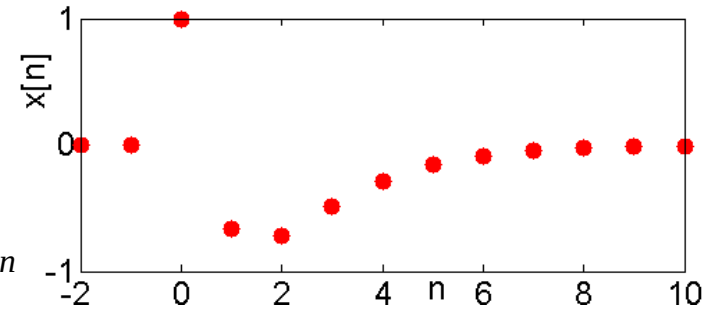
$$= 7 \sum_{n=0}^{\infty} (1/3)^n z^{-n} - 6 \sum_{n=0}^{\infty} (1/2)^n z^{-n}$$

$$= \frac{7z}{z - 1/3} - \frac{6z}{z - 1/2}$$

$$= \frac{z(z - 3/2)}{(z - 1/3)(z - 1/2)}$$

For the region of convergence we require both summations to converge $|z| > 1/3$ and $|z| > 1/2$, so

$$|z| > 1/2$$



Lecture 18: Summary

The z-transform can be used to represent discrete-time signals for which the discrete-time Fourier transform does not converge

It is given by:

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

where z is a complex number. The aim is to represent a discrete time signal in terms of the basis functions (z^n) which are subject to a magnitude and phase shift when processed by a discrete time system.

The z-transform has an associated region of convergence for z , which is determined by when the infinite sum converges. Often $X(z)$ is evaluated using an infinite sum.

Lecture 18: Exercises

Theory

SaS O&W: 10.1-10.4

Matlab

You can use the `ztrans()` function which is part of the symbolic toolbox. It evaluates signals $x[n]u[n]$, i.e. for non-negative values of n .

```
syms k n w z
ztrans(2^n)
% returns z/(z-2)
ztrans(0.5^n)
% returns z/(z-0.5)
ztrans(sin(k*n),w)
% returns sin(k)*w/(1*w*cos(k)+w^2)
```

Note that there is also the `iztrans()` function (see next lecture)